room). It is the combination of data acquisition, processing, storage, contextualization, and visualization that makes informative digital twins so valuable.

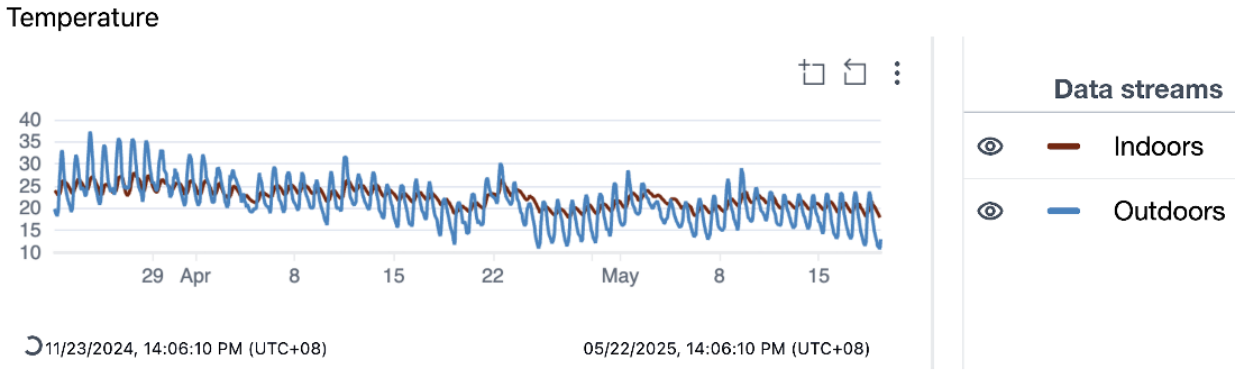


Figure 1.7 Information about the physical environment represented in a dashboard forms the basis of many informative digital twins.

1.3.3 Predictive digital twin

A *predictive digital twin* forecasts what the state of the physical system might be in the future and does not only reflect its current state. It uses historical data to anticipate future states of a physical system, based on the assumption that future states and behavior will be consistent with historical patterns. These predictions can range from simple rules, like alerting when a temperature reading crosses a dangerous threshold, to complex machine learning models trained on months or years of operational data.

Predictive digital twins are particularly valuable in industries that require extensive and costly maintenance of equipment, as they can help them move from periodic (replace parts every 6 months) or reactive (replace parts when the machine breaks) maintenance to predictive maintenance—replacing parts when they start to exhibit characteristics that may indicate failure.